

1. Which of the following cannot form a set? ()
- A. All positive numbers B. Numbers equal to 2 C. Numbers close to 0
- D. Even numbers not equal to 0
2. Given that $x \in \mathbb{N}$, the solution set of the equation $x^2 + x - 2 = 0$ is ()
- A. $\{x | x = 2\}$ B. $\{1, -2\}$
- C. $\{x | x = 1\}$ D. $\{-1, 2\}$
3. The set equivalent to $M = \{x | x^2 - 9 = 0\}$ is ()
- A. $\{(-3, 3)\}$ B. $\{-3, 3\}$ C. $\{x = -3, x = 3\}$ D. $[-3, 3]$
4. The correct representation of set $\{x \in \mathbb{N} | x < 5\}$ using enumeration is ()
- A. $\{0, 1, 2, 3, 4\}$ B. $\{1, 2, 3, 4\}$ C. $\{0, 1, 2, 3, 4, 5\}$ D. $\{1, 2, 3, 4, 5\}$
5. The correct set representation using enumeration for the set $\{x | -3 \leq x \leq 3 \text{ and } x \in \mathbb{N}\}$ is ()
- A. $\{-3, -2, -1, 0, 1, 2, 3\}$ B. $\{-2, -1, 0, 1, 2\}$ C. $\{0, 1, 2, 3\}$ D. $\{1, 2, 3\}$
6. Given the universal sets $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2\}$, and $\complement_U A =$ (),
- A. $\{1, 2, 3\}$ B. $\{2, 3, 4\}$ C. $\{3, 4, 5\}$ D. $\{4, 5\}$
7. Given sets $A = \{x | -1 < x \leq 2\}$ and $B = \{-1, 0, 1, 2, 3\}$, then $A \cap B =$ ()
- A. $\{-1, 0, 1, 2\}$ B. $\{0, 1, 2\}$
- C. $\{0, 1\}$ D. $\{x | -1 < x \leq 2 \text{ or } x = 3\}$
8. The solution set for the inequality $(x + 5)(3 - 2x) \geq 6$ is ()
- A. $\left\{x \mid -\frac{9}{2} \leq x \leq 1\right\}$ B. $\left\{x \mid -1 < x < \frac{9}{2}\right\}$
- C. $\left\{x \mid x \leq -\frac{9}{2} \text{ or } x \geq 1\right\}$ D. $\left\{x \mid x \leq -1 \text{ or } x \geq \frac{9}{2}\right\}$
9. The solution set for the inequality $x^2 - 2x > 0$ is ()
- A. $\{x | x > 0\}$ B. $\{x | x < 2\}$ C. $\{x | 0 < x < 2\}$ D. $\{x | x < 0 \text{ or } x > 2\}$

10. The solution set for the inequality $\frac{3x-1}{2-x} \geq 1$ is ()

A. $\left\{x \mid \frac{3}{4} \leq x \leq 2\right\}$

B. $\left\{x \mid \frac{3}{4} \leq x < 2\right\}$

C. $\{x \mid x \leq -\frac{3}{4} \text{ or } x > 2\}$

D. $\{x \mid x < 2\}$

11. If the function $f(x) = (2a-1)x$ (where a is a real number) is decreasing on $\mathbb{R}$, then ()

A. $a \geq \frac{1}{2}$

B. $a \leq \frac{1}{2}$

C. $a > \frac{1}{2}$

D. $a < \frac{1}{2}$

12. Given the function $f(x) = \frac{x^2}{1+|x-1|}$, then $f(-2) =$ ()

A. -1

B. 0

C. 1

D. 2

13. The domain of the function $f(x) = \sqrt{1-2x}$ is ()

A. $\left(-\infty, \frac{1}{2}\right]$

B. $\left[\frac{1}{2}, +\infty\right)$

C. $(-\infty, 2]$

D. $[2, +\infty)$

14. The inverse function of $f(x) = \sqrt[3]{x-1}$ is $f^{-1}(x)$, which equals ()

A. $x^3 + 1$

B. $-x^3 + 1$

C. $x^{-3} + 1$

D. $x^3 - 1$

15. The inverse function of $y = \sqrt{x+1} (x \geq -1)$ is ()

A. $y = x^2 - 1 (x \geq 0)$

B. $y = x^2 - 1 (x \geq 1)$

C. $y = x^2 + 1 (x \geq 0)$

D. $y = x^2 + 1 (x \geq 1)$

16. The range of the function $y = \frac{6}{x-1}$ over the interval $[3, 4]$ is ()

A. $[1, 2]$

B. $[3, 4]$

C. $[2, 3]$

D. $[1, 6]$

17. Given that $f(x) = x^2 + 2x + 3$ is a monotonic function on $(-9, a)$, the range of a is ()

A. $(-\infty, -1)$

B. $(-\infty, -1]$

C. $(-9, -1)$

D. $(-9, -1]$

18. Let a , b , and c be real numbers such that $a > b > 0$. Which of the following inequalities holds? ()

A. $a^2 < b^2$

B. $ac^2 < bc^2$

C. $\frac{1}{a} < \frac{1}{b}$

D. $\frac{c}{a} < \frac{c}{b}$

19. Given that the graph of the power function $f(x)$ passes through the point $\left(2, \frac{1}{2}\right)$, the value of $f(4)$ is ()

- A. $\frac{1}{2}$ B. 2 C. 4 D. $\frac{1}{4}$

20. Which of the following calculations is correct? ()

- A. $(a^2)^3 = a^5$
 B. $\log_2 6 - \log_2 3 = 1$
 C. $a^{-\frac{1}{2}} \cdot a^{\frac{1}{2}} = 0$
 D. $\log_3 (-4)^2 = 2 \log_3 (-4)$

21. If $\sin \theta + \cos \theta = \frac{7}{5}$, then $\sin \theta \cos \theta =$ ()

- A. $\frac{24}{25}$ B. $\frac{12}{25}$
 C. $\pm \frac{24}{25}$ D. $-\frac{24}{25}$

22. The value of $\tan \frac{8\pi}{3}$ is ()

- A. $\frac{\sqrt{3}}{3}$ B. $-\frac{\sqrt{3}}{3}$ C. $\sqrt{3}$ D. $-\sqrt{3}$

23. Given that $\alpha \in \left(-\frac{\pi}{2}, 0\right)$ and $\sin \alpha = -\frac{3}{5}$, then $\tan \frac{\alpha}{2} =$ ()

- A. 3 B. -3 C. $\frac{1}{3}$ D. $-\frac{1}{3}$

24. Given that the general term formula of the sequence $\{a_n\}$ is $a_n = n^2 + 2n$, then $a_{10} =$ ()

- A. 100 B. 110 C. 120 D. 130

25. Given the sequence $\{a_n\}$ satisfies $a_{n+1} = 1 + \frac{1}{a_n}$ and $a_4 = \frac{8}{5}$, then $a_1 =$ ()

- A. $\frac{2}{3}$ B. 1 C. $\frac{5}{3}$ D. 2

26. Given the arithmetic sequence $\{a_n\}$, where the sum of the *first* n terms is S_n ,

if $a_2 + a_3 + a_{14} + a_{15} = 40$, then $S_{16} =$ ()

- A. 150 B. 160 C. 170 D. 180

27. In the geometric sequence $\{a_n\}$, given that $a_2 = 4$ and $a_5 = 32$, then $a_4 =$ ()

- A. 8 B. -8 C. 16 D. -16

28. Among the following curves, the slope of the tangent line at $x = 1$ is $\frac{3\pi}{4}$. ()

A. $y = x^2 - \frac{3}{x}$

B. $y = x \ln x$

C. $y = x^2 - 1$

D. $y = x^3 - 2x^2$

29. Given that line $l_1 : ax - 2y + 3 = 0$ intersects line $l_2 : x + (a - 3)y + 1 = 0$, if $l_1 \perp l_2$, then $a =$ ()

A. 6

B. -6

C. 2

D. -2

30. The coordinates of the intersection point between line $x = 2$ and line $y = x + 1$ are ()

A. (2, 3)

B. (-2, -3)

C. (0, 1)

D. (0, 0)

31. The standard equation of the circle with center $(2, -5)$ and radius 4 is ()

A. $(x + 2)^2 + (y - 5)^2 = 16$

B. $(x - 2)^2 + (y + 5)^2 = 16$

C. $(x + 2)^2 + (y - 5)^2 = 4$

D. $(x - 2)^2 + (y + 5)^2 = 4$

32. If the focal length of the ellipse $x^2 + my^2 = 1$ is 2, then the value of m is ()

A. $\frac{1}{2}$

B. 1

C. 2

D. 4

33. The coordinates of the foci of the hyperbola $\frac{y^2}{9} - \frac{x^2}{16} = 1$ are ()

A. $(\pm 5, 0)$

B. $(0, \pm 5)$

C. $(\pm\sqrt{5}, 0)$

D. $(0, \pm\sqrt{5})$

34. The directrix equation of the parabola $y^2 = -8x$ is ()

A. $y = -2$

B. $x = -4$

C. $x = 2$

D. $x = -2$

35. In parallelogram ABCD, $\overrightarrow{AB} + \overrightarrow{BD} - \overrightarrow{AC} =$ ()

A. $\overrightarrow{DC}$

B. $\overrightarrow{BA}$

C. $\overrightarrow{BC}$

D. $\overrightarrow{BD}$

36. Given $A(1, 2, 3)$ and $B(4, 5, 6)$, the coordinates of vector $\overrightarrow{AB}$ are ()

A. (3, 3, 3)

B. (5, 7, 9)

C. (-3, -3, -3)

D. (3, 7, 3)

37. Given vectors $\vec{a} = (1, 2, -\sqrt{3})$ and $\vec{b} = (1, 1, \sqrt{3})$, then $\vec{a} \cdot \vec{b} =$ ()

A. -1

B. 0

C. 1

D. 2

38. Given vectors $\vec{a} = (2, 3, 1)$ and $\vec{b} = (1, 2, 0)$, then $|\vec{a} + \vec{b}|$ equals ()

- A. $\sqrt{3}$ B. 3 C. $\sqrt{35}$ D. 9

39. Given that $\vec{a} = (-1, 3, -2)$, $\vec{b} = (1, -1, m)$ and $\vec{a} \cdot \vec{b} = -2$, then $m =$ ()

- A. -1 B. 1 C. -2 D. 2

40. Given that the complex number z satisfies $z(1+2i) = i(1+z)$, then $z =$ ()

- A. $\frac{1}{2} + \frac{1}{2}i$ B. $\frac{1}{2} - \frac{1}{2}i$ C. $1+i$ D. $1-i$

41. $(2+i) + (3+i) =$ ()

- A. $5+2i$ B. $5+5i$ C. $6+i$ D. $6+5i$

42. For the complex number $z = \frac{5i}{1+2i}$, then $|z| =$ ()

- A. 5 B. $5\sqrt{5}$ C. $2\sqrt{5}$ D. $\sqrt{5}$

43. Given that the complex number z satisfies $(1-i)z = 2i$, then $z =$ ()

- A. $-1+i$ B. $-1-i$ C. $1+i$ D. $1-i$

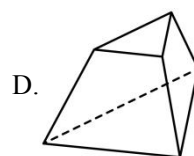
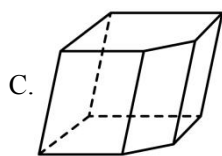
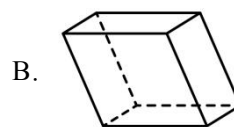
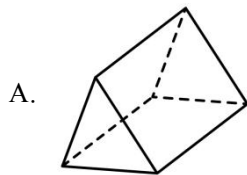
44. In the spatial Cartesian coordinate system $O-xyz$, the point $A(1, 2, 3)$ and the point $B(-1, -2, 3)$ ()

- A. Symmetrical about the origin B. Symmetrical with respect to the plane xOy
C. Symmetrical with respect to the plane yOz D. Symmetric about the axis z

45. The distance from point $P(1, -2, 5)$ to coordinate plane xOz is ()

- A. 2 B. 1 C. 5 D. 3

46. Which of the following solids is not a prism? ()



47. A box contains 2 red balls and 1 white ball, all of the same size. If one ball is randomly selected from the box, what is the probability of selecting the white ball? ()

- A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{2}{3}$ D. 1

48. Given that the normal density function of the random variable X

is $f(x) = \frac{1}{\sqrt{8\pi}} e^{-\frac{x^2}{8}} (x \in \mathbf{R})$, its mean and standard deviation are respectively ().

- A. 0 and 8 B. 0 and 4 C. 0 and 2 D. 0 and 1

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Reference Answer

Question Number	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	A	C	C	B	A	D	B
Question Number	11	12	13	14	15	16	17	18	19	20
Answer	D	C	A	A	A	C	D	C	D	B
Question Number	21	22	23	24	25	26	27	28	29	30
Answer	B	D	D	C	D	B	C	D	A	A
Question Number	31	32	33	34	35	36	37	38	39	40
Answer	B	A	B	C	B	A	B	C	A	A
Question Number	41	42	43	44	45	46	47	48	44	
Answer	A	D	A	D	A	D	A	C		

1. C

【Knowledge Point】 Determining whether elements can form a set

【Detailed Explanation】 Question Analysis: Elements in a set satisfy three criteria: definiteness, distinctness, and lack of order. "Numbers close to zero" are indefinite elements.

Therefore, numbers close to zero cannot form a set. Select C.

Test Point: Meaning of a set.

2. C

【Knowledge Point】 Equations and inequalities; representing sets using enumeration

【Analysis】 Solve the equation by factoring, then determine the solution set based on the problem context.

【Detailed Solution】 From the equation $x^2 + x - 2 = 0$, we obtain $(x+2)(x-1) = 0$. Solving yields $x = 1$ or $x = -2$.

Since $x \in \mathbb{N}$, we have $x = 1$. Thus, the solution set of the equation is $\{x \mid x = 1\}$.

Therefore, the correct answer is: C.

3. B

[Knowledge Point] Determining whether two sets are equal, representing sets using enumeration, representing sets using description

【Analysis】 Solve the equation to find the set M , then evaluate each option to determine its validity.

【Detailed Solution】 $M = \{x | x^2 - 9 = 0\} = \{-3, 3\}$,

Therefore, the set $\{-3, 3\}$ is equal to $M = \{x | x^2 - 9 = 0\}$. Thus, option B is correct.

Regarding A: $\{(-3, 3)\}$ is a point set, while M is a number set, so option A is incorrect;

For B: $\{x = -3, x = 3\}$ does not conform to the notation of set description, so option C is incorrect;

For D: $[-3, 3]$ represents real numbers greater than or equal to -3 and less than or equal to 3 , so option D is incorrect;

Therefore, the correct answer is: B.

4. A

[Knowledge Point] Set Notation

【Analysis】 From $x \in \mathbf{N}$, we know x is a natural number, then enumerate

【Detailed Explanation】 Since $x \in \mathbf{N}$ and $x < 5$, the values of x can be 0, 1, 2, 3, or 4.

Therefore, the correct answer is: A.

5. C

[Knowledge Point] Enumerative method represents sets; descriptive method represents sets.

【Analysis】 List elements satisfying the set conditions one by one to obtain the result.

【Detailed Explanation】 Since $\{x | -3 \leq x \leq 3\}$, and $x \in \mathbf{N} = \{0, 1, 2, 3\}$

Therefore, the correct answer is: C

【Key Insight】 This question tests the enumeration method and basic analytical problem-solving skills, classified as a foundational problem.

6. C

【Knowledge Point】 Concept and operations of complement sets

【Analysis】 Directly solve using the definition of complement.

【Detailed Solution】 Since the universal set is $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2\}$,

it follows that $\complement_U A = \{3, 4, 5\}$.

Therefore, the correct answer is: C

7. B

【Knowledge Point】 Concept and operation of intersection

【Analysis】 Solve using set intersection operations. >

【Detailed Solution】 Solution: $\because A = \{x \mid -1 < x \leq 2\}$, $B = \{-1, 0, 1, 2, 3\}$,

$\therefore A \cap B = \{0, 1, 2\}$.

Therefore, the correct answer is: B.

8. A

【Knowledge Point】 Solving Quadratic Inequalities Without Parameters

【Analysis】 When solving quadratic inequalities, if the right side of the inequality is not zero, first eliminate the non-zero term before solving.

【Detailed Solution】 The inequality $(x+5)(3-2x) \geq 6$ can be transformed into $2x^2 + 7x - 9 \leq 0$, which simplifies to $(x-1)(2x+9) \leq 0$. Solving this inequality yields $-\frac{9}{2} \leq x \leq 1$. Therefore, the solution set of this inequality is $\left\{x \mid -\frac{9}{2} \leq x \leq 1\right\}$.

Therefore, the correct answer is: A.

9. D

【Knowledge Point】 Solving Univariate Quadratic Inequalities Without Parameters

【Analysis】 Solve the quadratic inequality to obtain the solution.

【Detailed Solution】 Solve the inequality $x^2 - 2x > 0$, which is equivalent to $x(x-2) > 0$. Solving yields $x < 0$ or $x > 2$.

Therefore, the solution set of the original inequality is $\{x \mid x < 0 \text{ or } x > 2\}$.

Therefore, the correct answer is: D

10. B

【Knowledge Point】 Fractional Inequalities

【Analysis】 Convert the fractional inequality into a quadratic inequality in one variable for

solution.

【Detailed Solution】 Solution: Since $\frac{3x-1}{2-x} \geq 1$, $\frac{3x-1}{2-x} - 1 \geq 0$

$$\therefore \frac{3x-1-2+x}{2-x} \geq 0, \text{ i.e., } \frac{4x-3}{x-2} \leq 0, \therefore \begin{cases} (4x-3)(x-2) \leq 0 \\ x-2 \neq 0 \end{cases}, \text{ yielding } \frac{3}{4} \leq x < 2$$

Therefore, the correct answer is: B

11. D

[Knowledge Point] Functions, Graphs and Properties of Linear Functions

【Analysis】 The linear function is monotonically decreasing when $k < 0$

【Detailed Solution】 Given that $2a-1 < 0$, solving yields $a < \frac{1}{2}$

Therefore, the correct answer is: D

12. C

【Knowledge Point】 Finding Function Values

【Analysis】 Directly substitute $x = -2$ into the analytical expression to find the function value.

【Detailed Solution】 Given that $f(-2) = \frac{(-2)^2}{1+|-2-1|} = 1$.

Therefore, the correct answer is: C

【Key Insight】 This question tests the calculation of function values and is considered a foundational problem.

13. A

【Knowledge Point】 Domain of a specific function

【Analysis】 Solve the inequality to ensure the function is defined.

【Detailed Explanation】 For the function to be defined, we must have $1-2x \geq 0$, which implies $x \leq \frac{1}{2}$.

Thus, the domain of the function is $\left(-\infty, \frac{1}{2}\right]$.

Therefore, the correct answer is: A.

14. A

【Knowledge Point】 Finding the Inverse Function

【Analysis】 The solution can be derived using the definition of inverse functions.

【Detailed Solution】 Given that $y = \sqrt[3]{x-1}$,

Raising both sides to the third power yields: $y^3 = x - 1$

Simplifying yields: $x = y^3 + 1$

By the definition of inverse functions: $y = x^3 + 1$.

Therefore, the correct answer is: A.

15. A

[Knowledge Point] Finding the Inverse Function

[Analysis] From the original function, we can find $x = y^2 - 1$. By swapping x, y , we obtain the inverse function of the original function.

【Detailed Explanation】 Given $x \geq -1$, we obtain $y \geq 0$. Squaring both sides of $y = \sqrt{x+1}$ yields $y^2 = x+1$, i.e., $x = y^2 - 1$.

Swapping x and y yields $y = x^2 - 1$, whose domain is $[0, +\infty)$;

Therefore, the correct answer is: A.

16. C

[Knowledge Point] Directly determining the monotonicity of a function based on its analytical expression; common range values for functions (linear, quadratic, inverse proportional, etc.)

[Analysis] First determine the monotonicity of the function based on its analytical expression, then derive the range of the function within the given interval.

【Detailed Explanation】 Since the inverse proportional function $y = \frac{6}{x}$ is monotonically decreasing on $(0, +\infty)$, the graph of $y = \frac{6}{x-1}$ can be obtained by shifting the graph of $y = \frac{6}{x}$ one unit to the right.

Therefore, the function $y = \frac{6}{x-1}$ is monotonically decreasing on $(1, +\infty)$.

Therefore, $y = \frac{6}{x-1}$ is monotonically decreasing on the interval $[3, 4]$.

Therefore, $\frac{6}{4-1} \leq y \leq \frac{6}{3-1}$, i.e., $2 \leq y \leq 3$.

Therefore, the correct answer is: C.

17. D

[Knowledge Point] Determining the monotonicity of quadratic functions and finding monotonic

intervals; determining parameter values based on function monotonicity

【Analysis】 Determine the monotonicity of $f(x) = x^2 + 2x + 3$ to obtain $-9 < a \leq -1$.

【Detailed Explanation】 $f(x) = x^2 + 2x + 3$ is monotonically decreasing on $(-\infty, -1)$ and monotonically increasing on $(-1, +\infty)$. Therefore, for $(-9, a)$ to be a monotonic function, it must satisfy $-9 < a \leq -1$.

Therefore, the correct answer is: D

18. C

[Knowledge Point] Determine the validity of given inequalities based on known conditions

【Analysis】 By applying the properties of inequalities, we can determine the validity of each option to arrive at the correct answer.

【Detailed Explanation】 For Option A: If $a > b > 0$, then $a^2 > b^2$, so Option A is incorrect;

For option B: If $a > b > 0$, then $ac^2 \geq bc^2$, so option B is incorrect;

For option C: If $a > b > 0$, then $0 < \frac{1}{a} < \frac{1}{b}$, so option C is correct;

For option D: If $a > b > 0$, then $0 < \frac{1}{a} < \frac{1}{b}$. If $c \leq 0$, then $\frac{c}{a} \geq \frac{c}{b}$. Therefore, option D is incorrect.

Therefore, the correct answer is: C

19. D

[Knowledge Point] Finding the analytical expression of a power function, finding the value of a power function

[Analysis] Let the power function be $f(x) = x^a$. Substitute $(2, \frac{1}{2})$ into this equation to derive the analytical expression of the function, yielding the answer.

【Detailed Solution】 Let the power function be $f(x) = x^a$. The graph of the power function $f(x)$ passes through the point $(2, \frac{1}{2})$

Therefore, $f(2) = 2^a = \frac{1}{2}$. Solving yields $a = -1$

Therefore, $f(x) = x^{-1}$, so $f(4) = 4^{-1} = \frac{1}{4}$

Therefore, the correct answer is: D

20. B

[Knowledge Point] Operations with Exponential Powers, Operations with Logarithms

Analysis: Options A and C can be calculated using exponential operations; Option B can be calculated using logarithmic properties. Since the argument of a logarithm must be greater than zero, Option D can be determined accordingly.

【Detailed Explanation】 For A, $(a^2)^3 = a^6$, A is incorrect;

For B, $\log_2 6 - \log_2 3 = \log_2 \frac{6}{3} = \log_2 2 = 1$, B is correct;

For C, $a^{-\frac{1}{2}} \cdot a^{\frac{1}{2}} = a^0 = 1$, C is incorrect;

For D, since the argument of a logarithm must be greater than 0, D is incorrect.

Therefore, the correct answer is: B

21. B

[Knowledge Point] Relationship between $\sin \alpha \pm \cos \alpha$ and $\sin \alpha \cdot \cos \alpha$

【Analysis】 Square both sides of the given condition, simplify using the basic trigonometric identities for the same angle, and solve for $\sin \theta \cos \theta$.

【Detailed Solution】 Squaring both sides of $\sin \theta + \cos \theta = \frac{7}{5}$

yields $\sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta = \frac{49}{25}$, which simplifies to $1 + 2 \sin \theta \cos \theta = \frac{49}{25}$. Solving

for $\sin \theta \cos \theta = \frac{12}{25}$.

Therefore, the correct answer is B.

【Key Insight】 This question primarily tests the fundamental trigonometric identities for the same angle, assessing computational problem-solving skills. It is considered a foundational problem.

22. D

[Knowledge Point] Induction Formulas II, III, IV

【Detailed Solution】 $\tan \frac{8\pi}{3} = \tan \left(\frac{2\pi}{3} + 2\pi \right) = \tan \frac{2\pi}{3} = -\sqrt{3}$

23. D

【Knowledge Point】 Half-angle formula; finding cosine (or sine) given sine (or cosine)

【Analysis】 Derive $\cos \alpha = \frac{4}{5}$ from the angle's range and square relationship. Apply the

half-angle formula to find the target expression.

【Detailed Solution】 From $\tan \frac{\alpha}{2} = \frac{\sin \alpha}{1 + \cos \alpha}$, and since $\alpha \in \left(-\frac{\pi}{2}, 0\right)$ and $\sin \alpha = -\frac{3}{5}$, we have $\cos \alpha = \frac{4}{5}$.

Therefore, $\tan \frac{\alpha}{2} = -\frac{1}{3}$.

Therefore, the correct answer is: D

24. C

[Knowledge Point] Determining or writing terms in a sequence

【Analysis】 In the general term formula of the sequence $\{a_n\}$, let $n = 10$, then we can obtain the value of a_{10} .

【Detailed Explanation】 The general term formula for the sequence $\{a_n\}$ is $a_n = n^2 + 2n$, then $a_{10} = 10^2 + 2 \times 10 = 120$.

Therefore, the correct answer is: C.

【Key Insight】 This question tests the ability to find a term in a sequence given its general term formula, using substitution to solve. It is a foundational problem.

25. D

[Knowledge Point] Derive sequence terms based on recursive formulas

【Analysis】 Given the value of a_4 and the recursive formula, sequentially calculate a_3, a_2, a_1 to obtain the solution.

【Detailed Explanation】 Solution: Since $a_4 = 1 + \frac{1}{a_3} = \frac{8}{5}$, then $a_3 = \frac{5}{3}$. Furthermore, $a_3 = 1 + \frac{1}{a_2} = \frac{5}{3}$, so $a_2 = \frac{3}{2}$. Additionally, $a_2 = 1 + \frac{1}{a_1} = \frac{3}{2}$, hence $a_1 = 2$.

Therefore, the correct answer is: D.

26. B

[Knowledge Point] Basic calculation of the sum of the first n terms of an arithmetic sequence, finding the sum of the first n terms of an arithmetic sequence, calculation using properties of arithmetic sequences

【Analysis】 Calculate $a_2 + a_{15} = a_3 + a_{14} = 20$ based on the properties of arithmetic sequences,

then transform using the summation formula to obtain the answer.

【Detailed Explanation】 Since $\{a_n\}$ is an arithmetic sequence, $a_2 + a_{15} = a_3 + a_{14}$.

Since $a_2 + a_3 + a_{14} + a_{15} = 40$, we have $a_2 + a_{15} = a_3 + a_{14} = 20$.

$$S_{16} = \frac{16(a_1 + a_{16})}{2} = 8(a_2 + a_{15}) = 160.$$

Therefore, the correct answer is: B

27. C

【Knowledge Point】 Using the general term formula of a geometric sequence to find terms in the sequence; basic calculations of the general term formula for geometric sequences

【Explanation】 Calculate the common ratio of the geometric sequence based on the given conditions, then derive the value of a_4 using the transformed form of the geometric sequence's general term formula.

【Detailed Explanation】 Since $a_2 = 4, a_5 = 32$, we have $q^3 = \frac{a_5}{a_2} = 8$, so $q = 2$.

Therefore, $a_4 = a_2 q^2 = 4 \times 4 = 16$.

Therefore, the correct answer is: C.

28. D

【Knowledge Point】 Finding the equation (slope) of the tangent line at a point on a curve

【Analysis】 Differentiate the functions in the options and substitute $x = 1$ to determine.

【Detailed Solution】 Let $f(x) = x^3 - 2x^2$. Since $f'(x) = 3x^2 - 4x$, we

have $f'(1) = 3 - 4 = -1$. Thus, the slope of the tangent line to the curve $y = f(x)$ at $x = 1$ is $-\frac{3\pi}{4}$.

The other options do not satisfy this condition, so the correct answer is D.

【Key Insight】 This question tests the geometric meaning of derivatives and is a foundational problem.

29. A

【Knowledge Point】 Determining parameters when given perpendicular lines

【Analysis】 Derive the equation based on the necessary and sufficient condition for perpendicular lines, then solve the equation to obtain the answer.

【Detailed Solution】 Solution: Since line $l_1: ax - 2y + 3 = 0$ is perpendicular to

line $l_2 : x + (a-3)y + 1 = 0$ and $l_1 \perp l_2$,

it follows that $a \times 1 + (-2) \times (a-3) = 0$. Solving yields $a = 6$.

Therefore, the correct answer is: A.

30. A

[Knowledge Point] Finding the coordinates of the intersection point of two lines

【Analysis】 Directly solve the equation to find the coordinates of the intersection point of the two lines.

【Detailed Solution】 From $\begin{cases} x = 2 \\ y = x + 1 \end{cases}$, we solve for $\begin{cases} x = 2 \\ y = 3 \end{cases}$. Thus, the coordinates of the

intersection point between line $x = 2$ and line $y = x + 1$ are $(2, 3)$.

Therefore, the correct answer is: A.

31. B

[Knowledge Point] Finding the equation of a circle from its center (or radius), Finding the equation of a circle from its center (or radius)

【Analysis】 Directly write the result based on the center and radius.

【Detailed Explanation】 From the given conditions, the standard equation of the circle is $(x-2)^2 + (y+5)^2 = 16$.

Therefore, the correct answer is: B

32. A

【Knowledge Point】 Finding the foci and focal distance of an ellipse

【Analysis】 Solve using properties of ellipses.

【Detailed Solution】 $\because$ The ellipse $x^2 + my^2 = 1$ has a focal length of 2, and its standard equation is $x^2 + \frac{y^2}{m} = 1$.

Therefore, the foci cannot lie on the x axis. $\therefore 2\sqrt{\frac{1}{m}-1} = 2$, yielding $m = \frac{1}{2}$.

Therefore, the correct answer is: A.

【Key Insight】 This problem tests the use of an ellipse's focal length to determine parameters. As a

foundational question, careful reading and reasonable application of ellipse properties are essential.

33. B

[Knowledge Point] Finding the coordinates of the foci of a hyperbola

【Analysis】 First determine that the hyperbola's foci lie on the y axis, then calculate $c^2 = a^2 + b^2$.

[Detailed Solution] The foci of the hyperbola $\frac{y^2}{9} - \frac{x^2}{16} = 1$ lie on the y axis.

$$\because a^2 = 9, \quad b^2 = 16, \quad \therefore c^2 = a^2 + b^2 = 9 + 16 = 25,$$

$$\therefore c = 5,$$

The coordinates of the hyperbola's foci are $(0, \pm 5)$.

Therefore, the correct answer is: B.

34. C

[Knowledge Point] Determining the focus or directrix from the equation of a parabola

[Analysis] The focus can be determined using the equation of the directrix.

【Detailed Solution】 For the parabola $y^2 = -8x$, $2p = 8$, so $p = 4$.

Thus, the directrix equation of the parabola is $x = \frac{p}{2}$, i.e., $x = 2$.

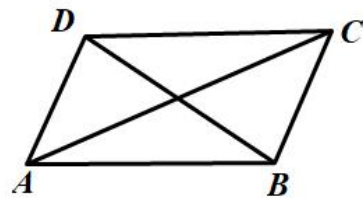
Therefore, the correct answer is: C

35. B

【Knowledge Point】 Rules for vector subtraction and addition

【Analysis】 The result can be derived using the rules for vector addition and subtraction.

【Detailed Solution】 In parallelogram $ABCD$,



$$\overrightarrow{AB} + \overrightarrow{BD} - \overrightarrow{AC} = \overrightarrow{AD} - \overrightarrow{AC} = \overrightarrow{CD} = \overrightarrow{BA}.$$

Therefore, the correct answer is: B.

36. A

【Knowledge Point】 Coordinate operations of spatial vectors

【Analysis】 Subtracting point A 's coordinates from point B 's yields:

【Detailed Explanation】 Since $A(1, 2, 3)$ and $B(4, 5, 6)$, we

have $\overrightarrow{AB} = (4, 5, 6) - (1, 2, 3) = (3, 3, 3)$.

Therefore, the correct answer is: A

37. B

【Knowledge Point】 Calculating the dot product of spatial vectors

【Analysis】 Calculate using the coordinate operations for the dot product of spatial vectors.

【Detailed Explanation】 Since $\vec{a} \cdot \vec{b} = 1 \times 1 + 2 \times 1 + (-\sqrt{3}) \times \sqrt{3} = 0$,

Therefore, the correct answer is: B

38. C

【Knowledge Point】 Coordinate representation of the magnitude of a spatial vector; coordinate operations of spatial vectors

【Analysis】 Calculate $\vec{a} + \vec{b}$ using the coordinate representation of spatial vector addition, then compute the magnitude using the formula for the magnitude of a spatial vector.

【Detailed Explanation】 $\vec{a} + \vec{b} = (3, 5, 1)$

Therefore $|\vec{a} + \vec{b}| = \sqrt{3^2 + 5^2 + 1^2} = \sqrt{35}$

Therefore, the correct answer is: C

39. A

【Knowledge Point】 Application of the dot product of spatial vectors

【Analysis】 Using the coordinate representation of the vector dot product, we calculate: $m = -1$.

【Detailed Solution】 From $\vec{a} = (-1, 3, -2)$, $\vec{b} = (1, -1, m)$, we obtain $\vec{a} \cdot \vec{b} = -1 - 3 - 2m = -2$.

Solving yields $m = -1$.

Therefore, the correct answer is: A

40. A

【Knowledge Point】 Division of complex numbers, multiplication of complex numbers in algebraic form

【Analysis】 From the problem statement, we have $z(1+i) = i$. Then, apply the arithmetic rules for complex numbers to calculate.

【Detailed Solution】 Solution: Since $z(1+2i) = i(1+z)$,

it follows that $z(1+i) = i$,

$$\text{Therefore, } z = \frac{i}{1+i} = \frac{i(1-i)}{(1+i)(1-i)} = \frac{1+i}{2} = \frac{1}{2} + \frac{1}{2}i,$$

Therefore, the correct answer is: A.

41. A

[Knowledge Point] Algebraic operations for complex number addition and subtraction

【Analysis】 Apply the addition rule.

$$[\text{Detailed Explanation}] (2+i) + (3+i) = 5+2i.$$

Therefore, the correct answer is: A

【Key Insight】 This question tests the addition of complex numbers and is a fundamental problem.

42. D

【Knowledge Point】 Finding the modulus of a complex number, multiplication of complex numbers in algebraic form, division of complex numbers

【Analysis】 Apply the four arithmetic operations of complex numbers to obtain $z = 2+i$, then solve using the modulus formula.

$$\text{【Detailed Solution】 } \because z = \frac{5i}{1+2i} = \frac{5i(1-2i)}{(1+2i)(1-2i)} = \frac{5i-10i^2}{5} = \frac{10+5i}{5} = 2+i,$$

$$\therefore |z| = \sqrt{2^2 + 1^2} = \sqrt{5}.$$

Therefore, the correct answer is: D.

43. A

【Knowledge Point】 Division of Complex Numbers

$$\text{【Analysis】 } (1-i)z = 2i$$

$$\text{【Detailed Explanation】 From } (1-i)z = 2i, \text{ we obtain } z = \frac{2i}{1-i} = i(1+i) = -1+i. \text{ Therefore,}$$

the correct answer is A.

[Test Focus] This question primarily assesses the four arithmetic operations with complex numbers. Complex numbers typically appear as short-answer questions in college entrance exams, classified as easy problems. The foundational content tested includes the concept of complex

numbers, their geometric meaning, and the four arithmetic operations.

44. D

【Knowledge Point】 Coordinates of points symmetric about the coordinate axes, coordinate planes, and origin

【Analysis】 Utilize the symmetry properties of points in space to determine the coordinates.

【Detailed Explanation】 In the Cartesian coordinate system, the coordinates of the point symmetrical to point (x, y, z) with respect to the x -axis are $(x, -y, z)$. Based on the given coordinates, we can determine that they are symmetrical with respect to the y -axis.

Therefore, the correct answer is: D

45. A

【Knowledge Point】 Position and Coordinate Characteristics of Points in Space

【Analysis】 Derived directly from the definition of spatial point coordinates.

【Detailed Explanation】 The distance from point $P(1, -2, 5)$ to coordinate plane xOz is $|-2| = 2$.

Therefore, the correct answer is: A.

46. D

【Knowledge Point】 Determining Whether a Solid is a Prism

【Analysis】 Solve based on the definition of a prism.

【Detailed Explanation】 According to the definition of a prism, A is a triangular prism, B is a quadrilateral prism, C is a pentagonal prism,

Only option D does not belong to the category of prisms.

Therefore, the correct answer is: D.

47. A

【Knowledge Point】 Calculating the probability of classical probability models

【Analysis】 Directly solve using the probability formula for classical probability models

【Detailed Solution】 Solution: Given that the box contains 3 balls of identical size—1 white ball and 2 red balls—

Therefore, the probability of randomly selecting a white ball is $\frac{1}{3}$,

Therefore, the correct answer is: A

【Key Insight】 This question tests the calculation of classical probability, representing a foundational problem.

48. C

【Knowledge Point】 Normal density function

【Analysis】 Directly solve using the meaning of the probability density function for a normal population.

【Detailed Explanation】 From the form $f(x)$, we know that the mean and standard deviation of X are 0 and 2, respectively.

The probability density function of the normal population is $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.

From $f(x) = \frac{1}{\sqrt{8\pi}} e^{-\frac{x^2}{8}} (x \in \mathbf{R})$, we find the mean is 0 and the standard deviation is 2.

Therefore, the correct answer is: C.